CHAPTER EXERCISE: Answers for this chapter start on page 299.

A calculator is allowed on the following questions.

1

Which of the following is a solution to the inequality $-x - 4 > 4x - 14$?

A) -1

B) 2

C) 5

D) 8

2

If $\frac{3}{4}x - 4 > \frac{1}{2}x - 10$, which of the following must be true?

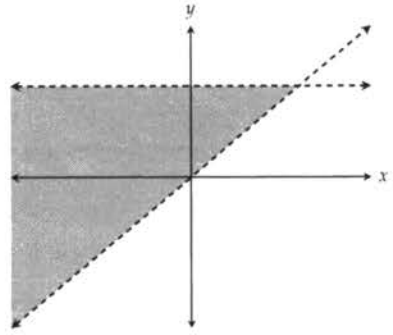
A) $x < 24$

B) $x > 24$

C) $x < -24$

D) $x > -24$

3



Which of the following systems of inequalities could be the one graphed in the xy-plane above?

A) $y > 3$
$y > x$

B) $y < 3$
$y < x$

C) $y < 3$
$y > x$

D) $y > 3$
$y < x$

4

Jerry estimates that there are m marbles in a jar. Harry, who knows the actual number of marbles in the jar, notes that the actual number, n, is within 10 marbles (inclusive) of Jerry's estimate. Which of the following inequalities represents the relationship between Jerry's estimate and the actual number of marbles in the jar?

A) $n + 10 \leq m \leq n - 10$

B) $m - 10 \leq n \leq m + 10$

C) $n \leq m \leq 10n$

D) $\frac{m}{10} \leq n \leq 10m$

5

A manufacturer produces chairs for a retail store according to the formula, $M = 12P + 100$, where M is the number of units produced and P is the retail price of each chair. The number of units sold by the retail store is given by $N = -3P + 970$, where N is the number of units sold and P is the retail price of each chair. What are all the values of P for which the number of units produced is greater than or equal to the number of units sold?

A) $P \geq 58$

B) $P \leq 58$

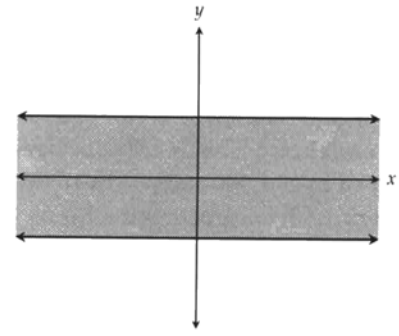
C) $P \geq 55$

D) $P \leq 55$

6

If n is an integer and $3(n-2) > -4(n-9)$, what is the least possible value of n ?

7



The graph in the xy-plane above could represent which of the following systems of inequalities?

A) $y \geq 3$
$y \leq -3$

B) $y \leq 3$
$y \geq -3$

C) $x \geq 3$
$x \leq -3$

D) $x \leq 3$
$x \geq -3$

8

To get to work, Harry must travel 8 miles by bus and 16 miles by train everyday. The bus travels at an average speed of x miles per hour and the train travels at an average speed of y miles per hour. If Harry's daily commute never takes more than 1 hour, which of the following inequalities represents the possible average speeds of the bus and train during the commute?

A) $\frac{8}{x} + \frac{16}{y} \leq 1$

B) $\frac{16}{x} + \frac{8}{y} \leq 1$

C) $\frac{x}{8} + \frac{y}{16} \leq 1$

D) $8x + 16y \leq 1$

9

An ice cream distributor contracts out to two different companies to manufacture cartons of ice cream. Company A can produce 80 cartons each hour and Company B can produce 140 cartons each hour. The distributor needs to fulfill an order of over 1,100 cartons in 10 hours of contract time. It contracts out x hours to Company A and the remaining hours to Company B. Which of the following inequalities gives all possible values of x in the context of this problem?

A) $\frac{80}{x} + \frac{140}{10 - x} > 1,100$

B) $140x + 80(10 - x) > 1,100$

C) $80x + 140(10 - x) > 1,100$

D) $80x + 140(x - 10) > 1,100$

10

$$y > 15x + a$$
$$y < 5x + b$$

In the system of inequalities above, a and b are constants. If $(1, 20)$ is a solution to the system, which of the following could be the value of $b - a$?

A) 6

B) 8

C) 10

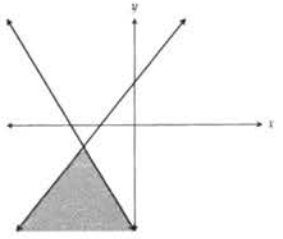
D) 12

11

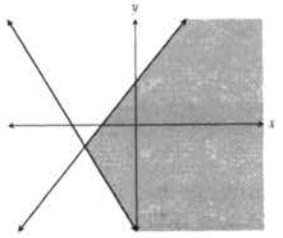
$$y \geq \frac{3}{2}x + 2$$
$$y \leq -2x - 5$$

Which of the following graphs in the xy-plane could represent the system of inequalities above?

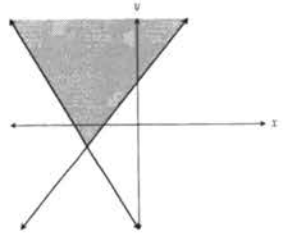
A)



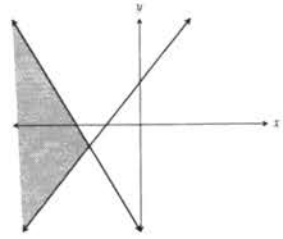
B)



C)



D)



12

Tina works no more than 30 hours at a nail salon each week. She can do a manicure in 20 minutes and a pedicure in 30 minutes. Each manicure earns her \$25 and each pedicure earns her \$40, and she must earn at least \$900 to cover her expenses. If during one week, she does enough manicures m and pedicures p to cover her expenses, which of the following systems of inequalities describes her working hours and her earnings?

A) $3m + 2p \leq 30$
$25m + 40p \geq 900$

B) $2m + 3p \leq 30$
$25m + 40p \geq 900$

C) $\frac{m}{3} + \frac{p}{2} \leq 30$
$25m + 40p \geq 900$

D) $\frac{m}{3} + \frac{p}{2} \geq 900$
$25m + 40p \leq 30$

13

If $k \leq x \leq 3k + 12$, which of the following must be true?

I. $x - 12 \leq 3k$
II. $k \geq -6$
III. $x - k \geq 0$

A) I only
B) I and II only
C) II and III only
D) I, II, and III

14

If $-\frac{20}{3} < -2x + 4 < -\frac{9}{2}$, what is one possible value of $x - 2$?

15

Joyce wants to create a rectangular garden that has an area of at least 300 square meters and a perimeter of at least 70 meters. If the length of the garden is x meters long and the width is y meters long, which of the following systems of inequalities represents Joyce's requirements?

A) $xy \geq 70$
$x + y \geq 300$

B) $xy \geq 150$
$x + y \geq 70$

C) $xy \geq 300$
$x + y \geq 70$

D) $xy \geq 300$
$x + y \geq 35$

16

If $a < b$, which of the following must be true?

I. $a^2 < b^2$
II. $2a < 2b$
III. $-b < -a$

A) II only
B) I and II only
C) II and III only
D) I, II, and III

Chapter 11: Inequalities

CHAPTER EXERCISE:

1. $\boxed{A}$

$$-x - 4 > 4x - 14$$
$$-5x > -10$$
$$x < 2$$

Of the answer choices, only -1 is a solution.

2. $\boxed{D}$ Multiply both sides by 4 to get rid of the fractions.

$$\frac{3}{4}x - 4 > \frac{1}{2}x - 10$$
$$3x - 16 > 2x - 40$$
$$x > -24$$

3. $\boxed{C}$ The shaded region falls below the horizontal line $y = 3$, so $y < 3$. The shaded region also stays above $y = x$, so $y > x$.

4. $\boxed{B}$ Let's say Jerry's estimate, m, is 100 marbles. If the actual number of marbles is within 10 of that estimate, then the actual number must be at least 90 and at most 110. Using variables, $m - 10 \leq n \leq m + 10$.

5. $\boxed{A}$ Setting up the inequality,

$$M \geq N$$
$$12P + 100 \geq -3P + 970$$
$$15P \geq 870$$
$$P \geq 58$$

6. $\boxed{7}$

$$3(n - 2) > -4(n - 9)$$
$$3n - 6 > -4n + 36$$
$$7n > 42$$
$$n > 6$$

Since n is an integer, the least possible value of n is 7.

7. $\boxed{B}$ The shaded region is below the horizontal line $y = 3$ but above the horizontal line $y = -3$. Therefore, $y \geq -3$ and $y \leq 3$.

8. $\boxed{A}$ The time Harry spends on the bus is $\frac{8}{x}$ hours and the time he spends on the train is $\frac{16}{y}$ hours. Since the total number of hours is never greater than 1, $\frac{8}{x} + \frac{16}{y} \leq 1$.

9. $\boxed{C}$ If the distributor contracts out to Company A for x hours, then it contracts out to Company B for $10 - x$ hours. Company A then produces $80x$ cartons and Company B produces $140(10 - x)$ cartons. Setting up the inequality,

$$80x + 140(10 - x) > 1,100$$

10. $\boxed{D}$ Plug in $x = 1$, $y = 20$ into the first inequality to get $20 > 15 + a$, $5 > a$. Do the same for the second inequality to get $20 < 5 + b$, $15 < b$. So, a is less than 5 and b is greater than 15. The difference between the two must be more than $15 - 5 = 10$. Among the answer choices, 12 is the only one that is greater than 10.

11. $\boxed{D}$ The line going from the bottom-left to the top-right must be $y = \frac{3}{2}x + 2$ and the line going from the top-left to the bottom-right must be $y = -2x + 5$ (based on the slopes and y-intercepts). Answer (D) correctly shades in the region above $y = \frac{3}{2}x + 2$ and below $y = -2x + 5$.

12. $\boxed{C}$ One manicure takes $1/3$ of an hour. One pedicure takes $1/2$ an hour. The total number of hours she spends doing manicures and pedicures must be less than or equal to 30, so $\frac{1}{3}m + \frac{1}{2}p \leq 30$. She earns $25m$ for the manicures and $40p$ for the pedicures. Altogether, $25m + 40p \geq 900$.

13. $\boxed{D}$ From the given inequality, $x \leq 3k + 12$. Subtracting 12 from both sides gives $x - 12 \leq 3k$, which confirms that I is always true.

From the given inequality, $3k + 12 \geq k$, which means $2k \geq -12$, $k \geq -6$, so II must also be true.

From the given inequality, $k \leq x$. Subtracting k from both sides gives $0 \leq x - k$. Therefore, III must also be true.

14. $\boxed{\frac{9}{4} < x < \frac{10}{3}}$ Let's solve these separately. First,

$$\begin{aligned} -\frac{20}{3} &< -2x + 4 \\ -20 &< -6x + 12 \\ -32 &< -6x \\ \frac{16}{3} &> x \end{aligned}$$

Now for the second part,

$$\begin{aligned} -2x + 4 &< -\frac{9}{2} \\ -4x + 8 &< -9 \\ -4x &< -17 \\ x &> \frac{17}{4} \end{aligned}$$

Putting the two results together, $\frac{17}{4} < x < \frac{16}{3}$. Therefore, $\frac{9}{4} < x - 2 < \frac{10}{3}$.

15. $\boxed{D}$ If the area is at least 300, then $xy \geq 300$. The perimeter of the rectangular garden is $2x + 2y$, so $2x + 2y \geq 70$, which reduces to $x + y \geq 35$.

16. $\boxed{C}$ I is not always true because of negative values. Take $a = -5$ and $b = 2$ for example. $a < b$, but $a^2 > b^2$. II is definitely true. It's the equivalent of multiplying both sides by 2. III is also true. It's the equivalent of multiplying both sides by -1, which necessitates a sign change.